Build physical intelligence.

A U.S. AI-driven hardware R&D and manufacturing company.

Self-developed joints. Cost-led engineering.
A repeatable path from design to production.

OCTOBER 2026

A real market. A focused entry point.

$2.25B

2025 robot order value

36,766 robots ordered in North America.
+10.1% order value year over year.

$1.166B

H1 2026 robot order value

17,995 robots ordered in North America.
+6.6% order value year over year.

Our entry market: U.S. robotics OEMs, applied-AI teams and automation integrators buying modular actuation and manipulation hardware.

A3 complete-robot order data. Different reporting periods; do not add them. These are not U.S.-only figures or FlatHill's component TAM. Q2 2026 non-automotive share: 56% of units.

A3 · 2025 North American robot orders · A3 · H1 / Q2 2026 robot orders

Three buyers. A clear first purchase.

01

Robotics OEMs

Paid joint evaluation kits → design-in engineering → scheduled component supply. Proof: accepted tests, released design and production purchase orders.

02

AI & research teams

Hand / arm development rigs, spares and integration support. Proof: an accepted experimental workflow and repeat purchases.

03

Automation integrators

Configured subsystems with a stable BOM and support boundary. Proof: site acceptance and replication across applications.

Focus on paid engineering needs. Convert accepted hardware into repeat orders.

Commercial strategy. No named customers, signed pipeline or order book is asserted.

One actuation foundation. Two scales.

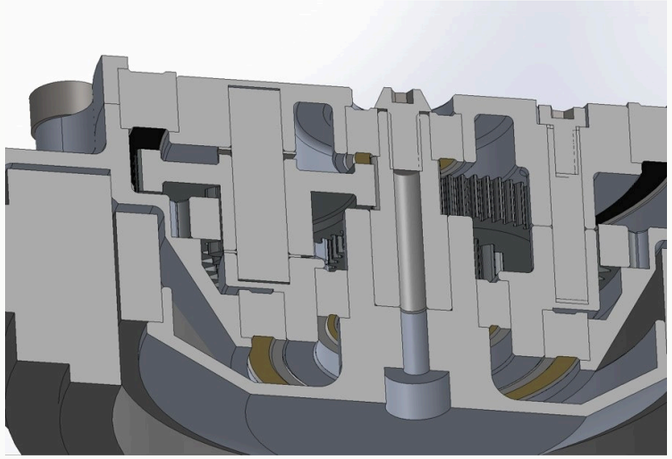


MICRO DRIVE FAMILY

M01 / M01 Neo / M02

Finger and end-effector architectures.

~$100 per module



LARGE JOINT FAMILY

WZ53 / WZ79 / WZ119

Joints for arms and larger mechanisms.

~$150–500 per module

Management-provided component budget inputs, not audited costs or selling prices. Self-developed module design can incorporate purchased motors and chips. Production maturity varies by SKU.

Company-supplied joint-module development plan and DexHand materials; management pricing inputs, Oct 2026.

Four applications. Shared engineering.



Quadruped

A wheel-leg platform built around modular joints, motion control and Jetson Orin compute.



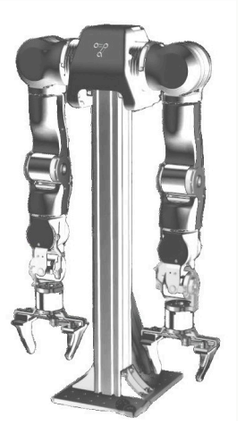
Dexterous hand

Miniature drives, replaceable fingers and documented interfaces for manipulation research and OEM integration.



Spherical robot

An enclosed drive and stabilization architecture with an AGX compute budget for sensing and autonomy development.



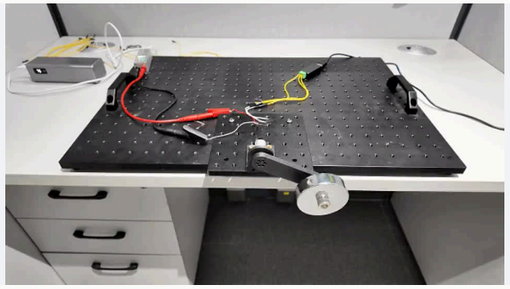
Dual-arm platform

Two seven-axis arms with a shared control and manufacturing architecture; configurable end effectors.

Product families demonstrate integration. Shared modules and production processes create the repeatable business.

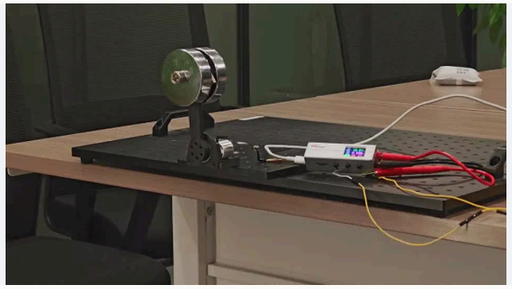
Existing company-supplied product renders. Configuration-specific qualification is required.

Watch the hardware move.



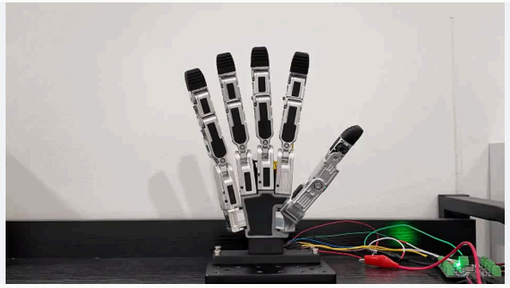
Micro drive · bench motion

Prototype test



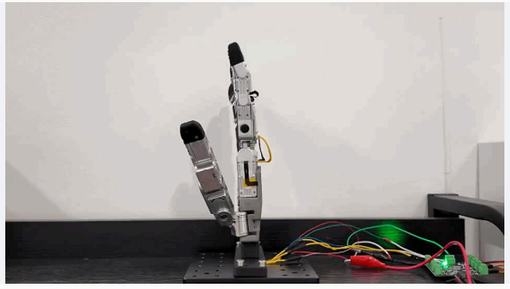
Micro drive · load fixture

Prototype test



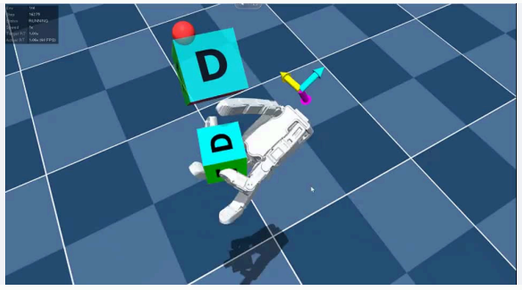
Dexterous hand · coordinated motion

Prototype demonstration



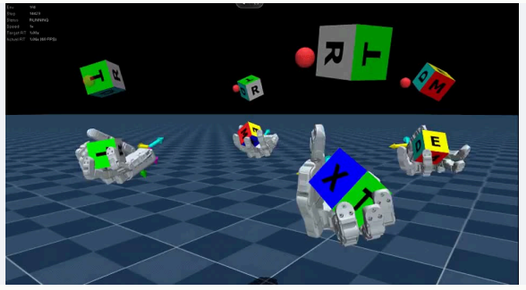
Dexterous hand · mechanism profile

Prototype demonstration



In-hand manipulation · simulation

Simulation



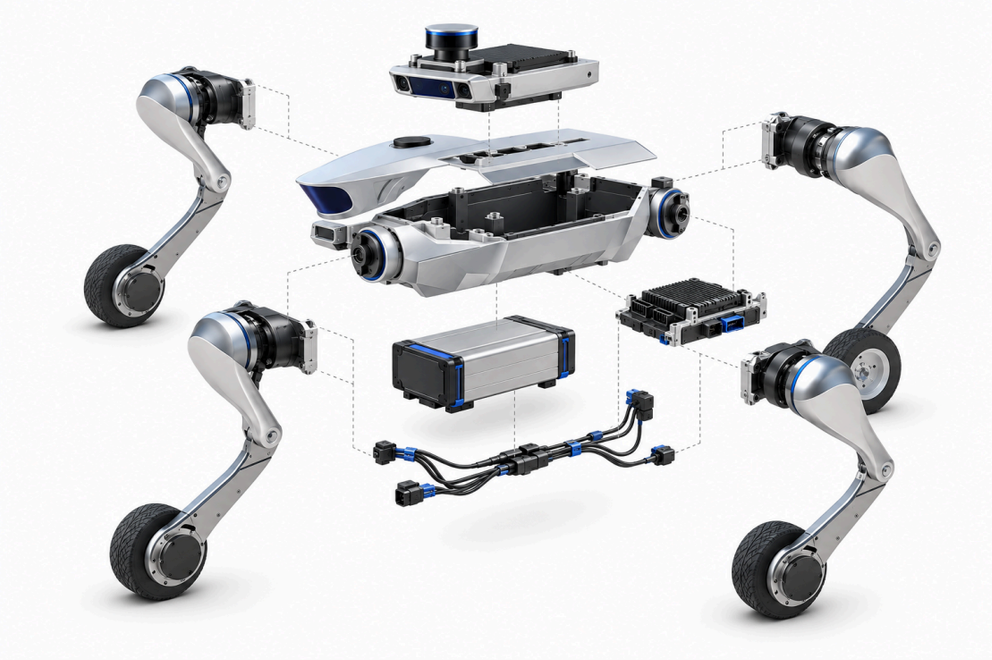
Parallel manipulation environments

Simulation

Open the website for eight company-supplied demonstration clips and two official NVIDIA technology references ↗

Qualitative prototype demonstrations and simulations; not independent performance validation. Third-party videos do not imply affiliation.

Quadruped: an inspectable cost model.



~$5.2k–$10.7k

Component budget, not selling price.
AI-assisted architecture illustration; not released CAD.

Component group	Budget / USD	Share*
Large joint modules	~$1,800–$6,000	~49%
Orin NX 16GB · production module	~$999	~13%
Compute integration	~$150–$300	~3%
CNC structures	~$800	~10%
Motion-control hardware	~$500	~6%
Four wheel drives	~$320–$600	~6%
Battery / BMS	~$350–$700	~7%
Sensing	~$200–$600	~5%
Harness / connectors	~$100–$200	~2%

*Midpoint shares, rounded. No origin qualification is inferred.
Planning configuration: 12 leg joints plus 4 separate wheel drives. Verify the released 16-axis configuration; wheel motors are not counted twice.

NVIDIA · Jetson pricing, lifecycle and origin · Joint unit inputs and dog CNC / control costs: management. Remaining quantities and allowances: planning assumptions.

Dexterous hand: an inspectable cost model.



Component group	Budget / USD	Share*
Micro drive modules	~$2,000	~71%
Finger / palm structures	~$200–$400	~11%
Control electronics	~$100–$200	~5%
Tactile / force sensing	~$150–$450	~11%
Wrist / harness	~$50–$100	~3%

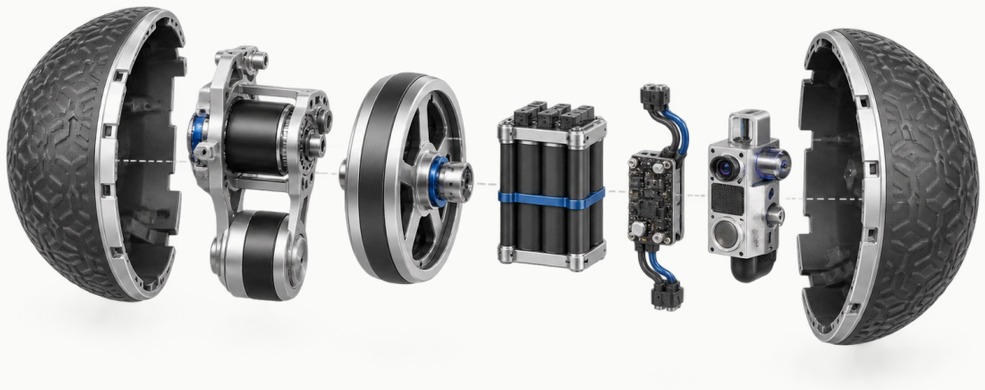
*Midpoint shares, rounded. No origin qualification is inferred.
Planning configuration: one hand with 20 independently driven joints at ~$100 each. Actuator count is an assumption, not inferred from DOF alone. External host excluded.

~$2.5k–$3.2k

Component budget, not selling price.
AI-assisted architecture illustration; not released CAD.

NVIDIA · Jetson pricing, lifecycle and origin · Joint unit inputs and dog CNC / control costs: management. Remaining quantities and allowances: planning assumptions.

Spherical robot: an inspectable cost model.



~$6.9k–$10.2k

Component budget, not selling price.
AI-assisted architecture illustration; not released CAD.

Component group	Budget / USD	Share*
AGX Orin 64GB · developer kit	~$3,499	~41%
Compute integration	~$50–$150	~1%
Shell / chassis	~$800–$1,500	~13%
Pendulum drive package	~$600–$1,200	~10%
Flywheel / stabilization	~$300–$700	~6%
Battery / BMS	~$800–$1,400	~13%
Motion / power control	~$400–$700	~6%
Sensing	~$300–$800	~6%
Harness / seals	~$150–$300	~3%

*Midpoint shares, rounded. No origin qualification is inferred.
AGX family specified by management; AGX Orin 64GB is the modeling assumption. Drive packages use budget allowances until motor count, power and supplier quotes are frozen.

NVIDIA · Jetson pricing, lifecycle and origin · Joint unit inputs and dog CNC / control costs: management. Remaining quantities and allowances: planning assumptions.

Dual-arm platform: an inspectable cost model.



~$10.1k–$18.2k

Component budget, not selling price.
AI-assisted architecture illustration; not released CAD.

Component group	Budget / USD	Share*
Large joint modules	~$2,100–$7,000	~32%
Micro drive modules	~$4,000	~28%
Orin NX 16GB · production module	~$999	~7%
Compute integration	~$150–$300	~2%
Two hands: non-drive parts	~$1,000–$2,300	~12%
Arm links / base	~$700–$1,400	~7%
Motion-control hardware	~$500–$800	~5%
Power supply	~$200–$400	~2%
Sensing	~$300–$700	~4%
Harness / connectors	~$150–$300	~2%

*Midpoint shares, rounded. No origin qualification is inferred.
Planning configuration: 14 arm joints + two 20-drive hands. Micro drives and hand structures are separate lines; no complete-hand cost is added again.

NVIDIA · Jetson pricing, lifecycle and origin · Joint unit inputs and dog CNC / control costs: management. Remaining quantities and allowances: planning assumptions.

A budget is not a compliance certificate.

01

Separate the cost models

Component budget excludes end-product assembly, test labor, yield loss, freight, duties, tax, warranty and operating expense. Validate landed COGS before setting margin.

02

Qualify origin by SKU

A U.S. supplier invoice or U.S. brand does not prove domestic origin. Unknown origin stays foreign; direct components are counted once. Imported joints do not change origin through resale.

03

Build the evidence file

General domestic test: U.S. manufacture plus component cost exceeding 65% through 2028; exceeding 75% from 2029. Exceptions and FCC authorization need separate review.

NVIDIA lists U.S.-origin module SKUs for Orin NX 16GB and AGX Orin 64GB. Obtain allocation, actual pricing and origin declarations for the purchased lots.

NVIDIA · Jetson pricing, lifecycle and origin · FAR 25.101 · Domestic end products · FAR 25.003 · Component-cost definitions

Earn the first order. Grow with the design win.

01

Evaluate

Hardware kit revenue

Paid evaluation with an agreed acceptance test.

02

Design in

Engineering + pilot lots

Released design, sourcing qualification and pilot acceptance.

03

Repeat

Scheduled supply

Production POs, traceability, stable quality and spares support.

04

Expand

ROSE + manufacturing

Paid workflow adoption and qualified U.S. production cells after 2027.

Factory capacity follows repeat demand and good-unit economics.

Proposed commercial model. Software subscriptions and manufacturing expansion are roadmap items.

Revenue follows customer adoption.

~$7.5M

5 programs × 500 robots / year × 12 modules × $250 selling price

~30,000 module shipments per year in one hypothetical production cohort.

Illustrative steady-state scenario, not current revenue, forecast, booked pipeline or TAM. The selling-price assumption is separate from component cost. No whole-robot, software or engineering revenue is added to these units. Gross margin requires validated landed cost, yield and support expense.

Adjust the model on the website ↗

Three stages. Evidence at every transition.

01

Now → 2027

Hardware qualification, paid evaluation kits, design wins and U.S. manufacturing partners. Gate: accepted pilots, repeat POs, costed BOMs and origin records.

02

After 2027 / 2028+

Commercialize ROSE workflows and dedicated U.S. assembly, calibration and test cells. Gate: paid workflow acceptance, stable yield and justified capacity.

03

Long-term / gated

Lights-out operations and controlled recursive improvement. Gate: validated changes, rollback, traceability and unattended reliability.

Development and commercialization roadmap. No booked sales pipeline or factory completion date is claimed.

Current evidence: supplied engineering documents and prototype media. ROSE early prototype is management-reported.

From intent to a manufacturable robot.

ROSE

DRAFTGPT WORKFLOW 01

From scratch to demo

Requirements → configuration → engineering review → tested prototype.

Output: design revisions, test results and acceptance record.

DRAFTGPT WORKFLOW 02

From demo to mass production

DFM → supplier qualification → process control → accepted pilot.

Output: manufacturing package, traceable build records and release decision.

An early prototype is reported by management. These workflows define the intended scope, not verified shipped functionality. Engineers approve design and manufacturing changes.

Planned commercialization after 2027. Interactive website explainer is not a live ROSE software demo.

Capacity follows proof.

01

Partner capacity

Qualify machining, assembly and pilot lots. Validate instructions, fixtures, acceptance tests and component records.

02

Dedicated U.S. cells

Bring critical assembly, calibration and test under direct process control when demand, yield and economics justify investment.

03

Lights-out operations

Automate stable repetitive tasks with monitoring, exception handling, recovery and traceability. Measure good-unit cost and reliability.

ROSE links the released design to production results. Recursive improvement remains bounded by engineering review and controlled release.

Proposed U.S. manufacturing roadmap. No existing lights-out factory or measured cost reduction is asserted.

Make compliance part of product release.

01

Classify the product

Apply the FCC’s defined advanced-robotic-device scope to each finished configuration. Fixed industrial arms and mobile systems need different analysis.

02

Document manufacture

Validate the U.S. manufacturing path and qualifying component cost. U.S.-supplier purchasing share is not the same measure. Normal equipment authorization is separate.

03

Establish rights

Verify IP assignments, third-party licenses, U.S. filing strategy and product-specific FTO. Financing history and patent registration do not independently prove eligibility.

Meta’s ESP32-based Muse SDK is a technology reference, not an exemption or a guarantee of future policy. FlatHill regulatory approvals and completed FTO are not asserted.

FCC · DA 26-786, Appendix C · FAR 25.101 · Domestic end products · USPTO · Managing a patent · Meta · Muse gadget SDK

Capital converts engineering into U.S. supply.

WORKING SEED PLAN

~$8–10M

September 2026 planning range.
Final round terms remain to be confirmed.

Prove a repeatable U.S. production program and validate ROSE on real work.

$10M ALLOCATION CASE

Factory / production equipment	20% · ~$2M
Team / operating runway	30% · ~$3M
R&D / materials / transfer	30% · ~$3M
Working capital / contingency	20% · ~$2M

At the lower end of the round, phase equipment and reserve use. Investor proof points: paid pilots, repeat orders, qualified BOMs, yield and accepted software workflows.

Existing management financing materials and allocation assumptions. No raised capital or committed investor is claimed.

Built by people who have shipped robots.

150+

robots delivered in
prior career programs

Management-reported patrol robotics experience.
Not FlatHill shipments or independently audited traction.

Hardware & actuation

Joint architecture, compact mechanisms, drive electronics and complete system integration.

Cost & manufacturing

Supplier coordination, pilot builds, process transfer, quality and field delivery.

Embedded control

Firmware, sensing and motion control tied to mechanical and electrical design.

Knowledge → ROSE

Reusable workflows, test records and controlled engineering changes.

Company-supplied team and product materials. Detailed references, responsibilities and IP assignments belong in diligence.

See the hardware.
Discuss the business.

A focused 30-minute investor session.

10 min	Product architecture & demonstrations
10 min	Commercial path & unit economics
10 min	Fundraising, milestones & diligence

flat-hill.github.io/flathill/#bookroadshow ↗

The website prepares an email meeting request. It does not confirm a calendar booking.